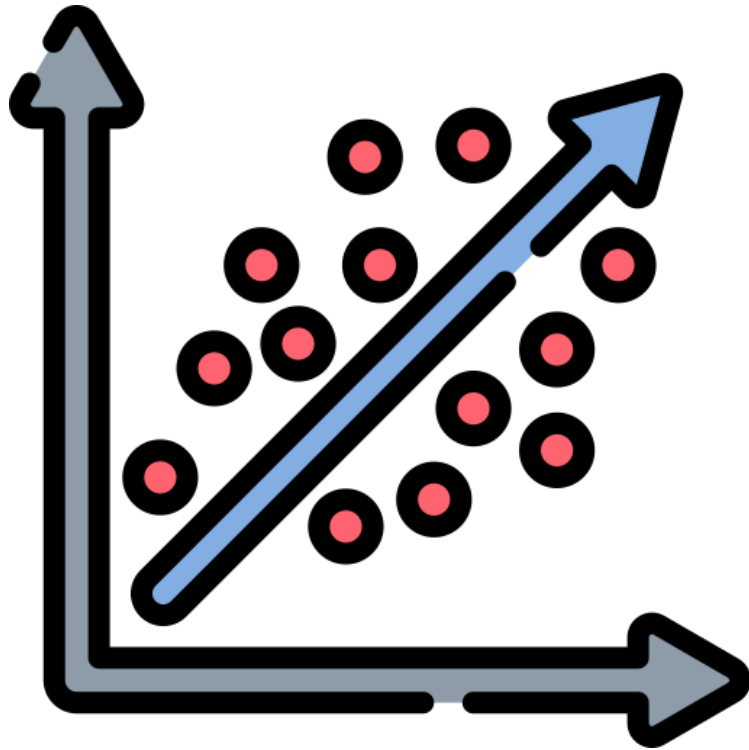


Regression Models for Sensor Prediction



Regression enables predictive insights from sensor data

What is Regression?

- 📈 Predicts continuous numerical values
- ✓ Outputs numbers, not categories

Common AIoT Applications

- 🌡️ Temperature trend prediction
- ⚡ Energy consumption forecasting
- 🕒 Remaining useful life (RUL) estimation
- 🏭 Equipment performance forecasting

How It Works

- ✓ Learns relationships between sensor inputs & future values
- ✓ Uses historical data to predict trends

Why Regression is Valuable in AIoT

- ✓ Simple & interpretable
- ✓ Computationally efficient
- ✓ Suitable for edge deployment
- ✓ Foundation for advanced models

Key Regression Models for Sensor Prediction



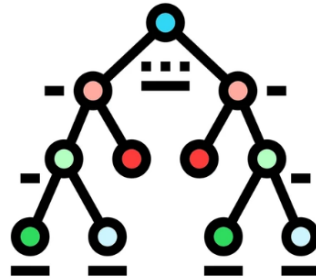
Linear & Multiple Linear Regression (LR / MLR)

- ✓ Best for linear relationships
✓ Simple, fast, and interpretable
- **Use for:** temperature, pressure, energy usage trends



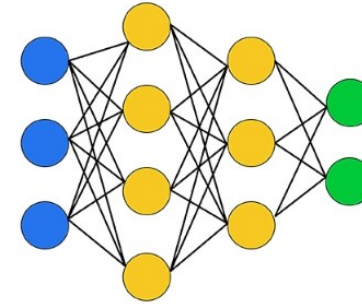
Partial Least Squares Regression (PLSR)

- ✓ Handles highly correlated sensor data
✓ Works well with multivariate inputs
- **Use for:** industrial sensors with overlapping signals



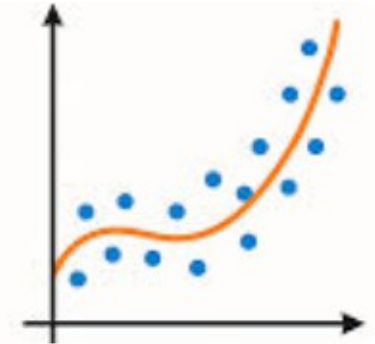
Decision Tree & Random Forest Regression

- ✓ Capture non-linear relationships
✓ Handle complex sensor interactions
- **Use for:** equipment performance & predictive maintenance



Deep Learning Regression (LSTM / CNN-LSTM)

- ✓ Designed for time-series forecasting
✓ Learns temporal patterns & sequences
- **Use for:** long-term demand, vibration trends, RUL prediction



Polynomial Regression

- ✓ Models curved & non-linear relationships
✓ Extends linear regression capability
- **Use for:** temperature variation patterns, load curves

Model choice depends on data complexity, time dependency, and computational constraints.

Common Applications and Techniques

Soft-Sensing

- Predict hard-to-measure parameters using available sensors
 - ✓ Example: pressure → wafer quality
 - ✓ Reduces need for expensive sensors

Data Cleaning & Preprocessing

- Regression blocks filter & smooth sensor data
 - ✓ Remove noise & outliers
 - ✓ Improve model accuracy
- *Example tools: Edge Impulse processing blocks*

Anomaly & Fault Detection

- Compare predicted vs actual values
 - ✓ Detect deviations & early faults
 - ✓ Enable predictive maintenance

Polynomial & Non-Linear Modeling

- Capture complex sensor behavior
 - ✓ Model curved & dynamic relationships
 - ✓ Improve accuracy in real environments

Regression techniques turn raw sensor signals into reliable insights

Common Applications and Techniques

Least-Squares Optimization

- ✓ Minimizes prediction error
- ✓ Finds best-fit line for sensor data
- ✓ Core method behind linear regression
- **Use:** improving accuracy of predictions

Data Transformation

- ✓ Fourier Transform → frequency patterns
- ✓ Wavelet Transform → time & frequency insights
- ✓ Enhances signal interpretation
- **Use:** vibration analysis, noise filtering, pattern detection

Feature Selection

- ✓ Select most relevant sensor inputs
- ✓ Removes redundant & noisy variables
- ✓ Improves model performance & efficiency
- **Use:** predictive maintenance, energy forecasting

Regression techniques turn raw sensor signals into reliable insights